

hw4_colab

March 7, 2025

```
[ ]: from google.colab import files  
files.upload()
```

```
[2]: !mkdir ~/.kaggle  
!cp kaggle.json ~/.kaggle/  
!chmod 600 ~/.kaggle/kaggle.json
```

```
[3]: !kaggle competitions download -c dogs-vs-cats
```

Downloading dogs-vs-cats.zip to /content
99% 804M/812M [00:04<00:00, 181MB/s]
100% 812M/812M [00:04<00:00, 196MB/s]

```
[4]: !unzip -qq dogs-vs-cats.zip
```

```
[5]: !unzip -qq train.zip
```

Copying images to training, validation, and test directories

```
[6]: import os, shutil, pathlib  
import tensorflow as tf  
from tensorflow import keras  
from keras import layers  
from keras.applications import ResNet50
```

```
[7]: new_base_dir = pathlib.Path("cats_vs_dogs_small")
```

```
[8]: original_dir = pathlib.Path("train")  
new_base_dir = pathlib.Path("cats_vs_dogs_small")  
  
def make_subset(subset_name, start_index, end_index):  
    for category in ("cat", "dog"):  
        dir = new_base_dir / subset_name / category  
        if os.path.exists(dir):  
            continue  
        os.makedirs(dir)  
        fnames = [f"{category}.{i}.jpg" for i in range(start_index, end_index)]  
        for fname in fnames:  
            shutil.copyfile(src=original_dir / fname,
```

```

dst=dir / fname)

make_subset("train", start_index=0, end_index=1000)
make_subset("validation", start_index=1000, end_index=1500)
make_subset("test", start_index=1500, end_index=2500)

```

Instantiating the ResNet50 convolutional base

```

[9]: conv_base = ResNet50(
    weights="imagenet", # Pre-trained on ImageNet
    include_top=False, # Exclude the fully-connected layer at the top
    input_shape=(180, 180, 3) # Input shape for your images
)

```

Downloading data from https://storage.googleapis.com/tensorflow/keras-applications/resnet/resnet50_weights_tf_dim_ordering_tf_kernels_notop.h5
94765736/94765736 1s
0us/step

```

[10]: conv_base.summary()

```

Model: "resnet50"

Layer (type)	Output Shape	Param #	Connected
↳ to			
input_layer (InputLayer)	(None, 180, 180, 3)	0	-
↳			
conv1_pad (ZeroPadding2D)	(None, 186, 186, 3)	0	↳
↳input_layer[0][0]			
conv1_conv (Conv2D)	(None, 90, 90, 64)	9,472	↳
↳conv1_pad[0][0]			
conv1_bn	(None, 90, 90, 64)	256	↳
↳conv1_conv[0][0]			
(BatchNormalization)			↳
↳			
conv1_relu (Activation)	(None, 90, 90, 64)	0	↳
↳conv1_bn[0][0]			
pool1_pad (ZeroPadding2D)	(None, 92, 92, 64)	0	↳
↳conv1_relu[0][0]			

pool1_pool (MaxPooling2D)	(None, 45, 45, 64)	0	┐
↳pool1_pad[0][0]			
conv2_block1_1_conv	(None, 45, 45, 64)	4,160	┐
↳pool1_pool[0][0]			
(Conv2D)			┐
↳			
conv2_block1_1_bn	(None, 45, 45, 64)	256	┐
↳conv2_block1_1_conv[0...			
(BatchNormalization)			┐
↳			
conv2_block1_1_relu	(None, 45, 45, 64)	0	┐
↳conv2_block1_1_bn[0][...			
(Activation)			┐
↳			
conv2_block1_2_conv	(None, 45, 45, 64)	36,928	┐
↳conv2_block1_1_relu[0...			
(Conv2D)			┐
↳			
conv2_block1_2_bn	(None, 45, 45, 64)	256	┐
↳conv2_block1_2_conv[0...			
(BatchNormalization)			┐
↳			
conv2_block1_2_relu	(None, 45, 45, 64)	0	┐
↳conv2_block1_2_bn[0][...			
(Activation)			┐
↳			
conv2_block1_0_conv	(None, 45, 45, 256)	16,640	┐
↳pool1_pool[0][0]			
(Conv2D)			┐
↳			
conv2_block1_3_conv	(None, 45, 45, 256)	16,640	┐
↳conv2_block1_2_relu[0...			
(Conv2D)			┐
↳			
conv2_block1_0_bn	(None, 45, 45, 256)	1,024	┐
↳conv2_block1_0_conv[0...			

(BatchNormalization)			└
↪			
conv2_block1_3_bn	(None, 45, 45, 256)	1,024	└
↪conv2_block1_3_conv[0...			
(BatchNormalization)			└
↪			
conv2_block1_add (Add)	(None, 45, 45, 256)	0	└
↪conv2_block1_0_bn[0][...			
			└
↪conv2_block1_3_bn[0][...			
conv2_block1_out	(None, 45, 45, 256)	0	└
↪conv2_block1_add[0][0]			
(Activation)			└
↪			
conv2_block2_1_conv	(None, 45, 45, 64)	16,448	└
↪conv2_block1_out[0][0]			
(Conv2D)			└
↪			
conv2_block2_1_bn	(None, 45, 45, 64)	256	└
↪conv2_block2_1_conv[0...			
(BatchNormalization)			└
↪			
conv2_block2_1_relu	(None, 45, 45, 64)	0	└
↪conv2_block2_1_bn[0][...			
(Activation)			└
↪			
conv2_block2_2_conv	(None, 45, 45, 64)	36,928	└
↪conv2_block2_1_relu[0...			
(Conv2D)			└
↪			
conv2_block2_2_bn	(None, 45, 45, 64)	256	└
↪conv2_block2_2_conv[0...			
(BatchNormalization)			└
↪			
conv2_block2_2_relu	(None, 45, 45, 64)	0	└
↪conv2_block2_2_bn[0][...			

(Activation)			└
↪			
conv2_block2_3_conv	(None, 45, 45, 256)	16,640	└
↪conv2_block2_2_relu[0...			
(Conv2D)			└
↪			
conv2_block2_3_bn	(None, 45, 45, 256)	1,024	└
↪conv2_block2_3_conv[0...			
(BatchNormalization)			└
↪			
conv2_block2_add (Add)	(None, 45, 45, 256)	0	└
↪conv2_block1_out[0][0...			
			└
↪conv2_block2_3_bn[0][...			
conv2_block2_out	(None, 45, 45, 256)	0	└
↪conv2_block2_add[0][0]			
(Activation)			└
↪			
conv2_block3_1_conv	(None, 45, 45, 64)	16,448	└
↪conv2_block2_out[0][0]			
(Conv2D)			└
↪			
conv2_block3_1_bn	(None, 45, 45, 64)	256	└
↪conv2_block3_1_conv[0...			
(BatchNormalization)			└
↪			
conv2_block3_1_relu	(None, 45, 45, 64)	0	└
↪conv2_block3_1_bn[0][...			
(Activation)			└
↪			
conv2_block3_2_conv	(None, 45, 45, 64)	36,928	└
↪conv2_block3_1_relu[0...			
(Conv2D)			└
↪			
conv2_block3_2_bn	(None, 45, 45, 64)	256	└
↪conv2_block3_2_conv[0...			

(BatchNormalization)				└
↪				
conv2_block3_2_relu	(None, 45, 45, 64)	0	└	
↪conv2_block3_2_bn[0][...]				
(Activation)				└
↪				
conv2_block3_3_conv	(None, 45, 45, 256)	16,640	└	
↪conv2_block3_2_relu[0...]				
(Conv2D)				└
↪				
conv2_block3_3_bn	(None, 45, 45, 256)	1,024	└	
↪conv2_block3_3_conv[0...]				
(BatchNormalization)				└
↪				
conv2_block3_add (Add)	(None, 45, 45, 256)	0	└	
↪conv2_block2_out[0][0...]				
			└	
↪conv2_block3_3_bn[0][...]				
conv2_block3_out	(None, 45, 45, 256)	0	└	
↪conv2_block3_add[0][0]				
(Activation)				└
↪				
conv3_block1_1_conv	(None, 23, 23, 128)	32,896	└	
↪conv2_block3_out[0][0]				
(Conv2D)				└
↪				
conv3_block1_1_bn	(None, 23, 23, 128)	512	└	
↪conv3_block1_1_conv[0...]				
(BatchNormalization)				└
↪				
conv3_block1_1_relu	(None, 23, 23, 128)	0	└	
↪conv3_block1_1_bn[0][...]				
(Activation)				└
↪				
conv3_block1_2_conv	(None, 23, 23, 128)	147,584	└	
↪conv3_block1_1_relu[0...]				

(Conv2D)			└
↪			
conv3_block1_2_bn	(None, 23, 23, 128)	512	└
↪conv3_block1_2_conv[0...			
(BatchNormalization)			└
↪			
conv3_block1_2_relu	(None, 23, 23, 128)	0	└
↪conv3_block1_2_bn[0] [...			
(Activation)			└
↪			
conv3_block1_0_conv	(None, 23, 23, 512)	131,584	└
↪conv2_block3_out[0] [0]			
(Conv2D)			└
↪			
conv3_block1_3_conv	(None, 23, 23, 512)	66,048	└
↪conv3_block1_2_relu[0...			
(Conv2D)			└
↪			
conv3_block1_0_bn	(None, 23, 23, 512)	2,048	└
↪conv3_block1_0_conv[0...			
(BatchNormalization)			└
↪			
conv3_block1_3_bn	(None, 23, 23, 512)	2,048	└
↪conv3_block1_3_conv[0...			
(BatchNormalization)			└
↪			
conv3_block1_add (Add)	(None, 23, 23, 512)	0	└
↪conv3_block1_0_bn[0] [...			
			└
↪conv3_block1_3_bn[0] [...			
conv3_block1_out	(None, 23, 23, 512)	0	└
↪conv3_block1_add[0] [0]			
(Activation)			└
↪			
conv3_block2_1_conv	(None, 23, 23, 128)	65,664	└
↪conv3_block1_out[0] [0]			

(Conv2D)			└
↪			
conv3_block2_1_bn	(None, 23, 23, 128)	512	└
↪conv3_block2_1_conv[0...			
(BatchNormalization)			└
↪			
conv3_block2_1_relu	(None, 23, 23, 128)	0	└
↪conv3_block2_1_bn[0][...			
(Activation)			└
↪			
conv3_block2_2_conv	(None, 23, 23, 128)	147,584	└
↪conv3_block2_1_relu[0...			
(Conv2D)			└
↪			
conv3_block2_2_bn	(None, 23, 23, 128)	512	└
↪conv3_block2_2_conv[0...			
(BatchNormalization)			└
↪			
conv3_block2_2_relu	(None, 23, 23, 128)	0	└
↪conv3_block2_2_bn[0][...			
(Activation)			└
↪			
conv3_block2_3_conv	(None, 23, 23, 512)	66,048	└
↪conv3_block2_2_relu[0...			
(Conv2D)			└
↪			
conv3_block2_3_bn	(None, 23, 23, 512)	2,048	└
↪conv3_block2_3_conv[0...			
(BatchNormalization)			└
↪			
conv3_block2_add (Add)	(None, 23, 23, 512)	0	└
↪conv3_block1_out[0][0...			
			└
↪conv3_block2_3_bn[0][...			
conv3_block2_out	(None, 23, 23, 512)	0	└
↪conv3_block2_add[0][0]			

(Activation)			└
↪			
conv3_block3_1_conv ↪conv3_block2_out[0][0] (Conv2D)	(None, 23, 23, 128)	65,664	└
↪			
conv3_block3_1_bn ↪conv3_block3_1_conv[0... (BatchNormalization)	(None, 23, 23, 128)	512	└
↪			
conv3_block3_1_relu ↪conv3_block3_1_bn[0][... (Activation)	(None, 23, 23, 128)	0	└
↪			
conv3_block3_2_conv ↪conv3_block3_1_relu[0... (Conv2D)	(None, 23, 23, 128)	147,584	└
↪			
conv3_block3_2_bn ↪conv3_block3_2_conv[0... (BatchNormalization)	(None, 23, 23, 128)	512	└
↪			
conv3_block3_2_relu ↪conv3_block3_2_bn[0][... (Activation)	(None, 23, 23, 128)	0	└
↪			
conv3_block3_3_conv ↪conv3_block3_2_relu[0... (Conv2D)	(None, 23, 23, 512)	66,048	└
↪			
conv3_block3_3_bn ↪conv3_block3_3_conv[0... (BatchNormalization)	(None, 23, 23, 512)	2,048	└
↪			
conv3_block3_add (Add) ↪conv3_block2_out[0][0...	(None, 23, 23, 512)	0	└

↪conv3_block3_3_bn[0][...]			└
conv3_block3_out	(None, 23, 23, 512)	0	└
↪conv3_block3_add[0][0]			
(Activation)			└
↪			
conv3_block4_1_conv	(None, 23, 23, 128)	65,664	└
↪conv3_block3_out[0][0]			
(Conv2D)			└
↪			
conv3_block4_1_bn	(None, 23, 23, 128)	512	└
↪conv3_block4_1_conv[0...			
(BatchNormalization)			└
↪			
conv3_block4_1_relu	(None, 23, 23, 128)	0	└
↪conv3_block4_1_bn[0][...]			
(Activation)			└
↪			
conv3_block4_2_conv	(None, 23, 23, 128)	147,584	└
↪conv3_block4_1_relu[0...			
(Conv2D)			└
↪			
conv3_block4_2_bn	(None, 23, 23, 128)	512	└
↪conv3_block4_2_conv[0...			
(BatchNormalization)			└
↪			
conv3_block4_2_relu	(None, 23, 23, 128)	0	└
↪conv3_block4_2_bn[0][...]			
(Activation)			└
↪			
conv3_block4_3_conv	(None, 23, 23, 512)	66,048	└
↪conv3_block4_2_relu[0...			
(Conv2D)			└
↪			
conv3_block4_3_bn	(None, 23, 23, 512)	2,048	└
↪conv3_block4_3_conv[0...			

(BatchNormalization)				└
↪				
conv3_block4_add (Add)	(None, 23, 23, 512)	0	└	
↪conv3_block3_out[0][0]				
↪conv3_block4_3_bn[0][...]			└	
conv3_block4_out	(None, 23, 23, 512)	0	└	
↪conv3_block4_add[0][0]				
(Activation)				└
↪				
conv4_block1_1_conv	(None, 12, 12, 256)	131,328	└	
↪conv3_block4_out[0][0]				
(Conv2D)				└
↪				
conv4_block1_1_bn	(None, 12, 12, 256)	1,024	└	
↪conv4_block1_1_conv[0]				
(BatchNormalization)				└
↪				
conv4_block1_1_relu	(None, 12, 12, 256)	0	└	
↪conv4_block1_1_bn[0][...]				
(Activation)				└
↪				
conv4_block1_2_conv	(None, 12, 12, 256)	590,080	└	
↪conv4_block1_1_relu[0]				
(Conv2D)				└
↪				
conv4_block1_2_bn	(None, 12, 12, 256)	1,024	└	
↪conv4_block1_2_conv[0]				
(BatchNormalization)				└
↪				
conv4_block1_2_relu	(None, 12, 12, 256)	0	└	
↪conv4_block1_2_bn[0][...]				
(Activation)				└
↪				
conv4_block1_0_conv	(None, 12, 12, 1024)	525,312	└	
↪conv3_block4_out[0][0]				

(Conv2D)			└
↪			
conv4_block1_3_conv	(None, 12, 12, 1024)	263,168	└
↪conv4_block1_2_relu[0...			
(Conv2D)			└
↪			
conv4_block1_0_bn	(None, 12, 12, 1024)	4,096	└
↪conv4_block1_0_conv[0...			
(BatchNormalization)			└
↪			
conv4_block1_3_bn	(None, 12, 12, 1024)	4,096	└
↪conv4_block1_3_conv[0...			
(BatchNormalization)			└
↪			
conv4_block1_add (Add)	(None, 12, 12, 1024)	0	└
↪conv4_block1_0_bn[0][...			
			└
↪conv4_block1_3_bn[0][...			
conv4_block1_out	(None, 12, 12, 1024)	0	└
↪conv4_block1_add[0][0]			
(Activation)			└
↪			
conv4_block2_1_conv	(None, 12, 12, 256)	262,400	└
↪conv4_block1_out[0][0]			
(Conv2D)			└
↪			
conv4_block2_1_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block2_1_conv[0...			
(BatchNormalization)			└
↪			
conv4_block2_1_relu	(None, 12, 12, 256)	0	└
↪conv4_block2_1_bn[0][...			
(Activation)			└
↪			
conv4_block2_2_conv	(None, 12, 12, 256)	590,080	└
↪conv4_block2_1_relu[0...			

(Conv2D)			└
↪			
conv4_block2_2_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block2_2_conv[0... (BatchNormalization)			└
↪			
conv4_block2_2_relu	(None, 12, 12, 256)	0	└
↪conv4_block2_2_bn[0] [... (Activation)			└
↪			
conv4_block2_3_conv	(None, 12, 12, 1024)	263,168	└
↪conv4_block2_2_relu[0... (Conv2D)			└
↪			
conv4_block2_3_bn	(None, 12, 12, 1024)	4,096	└
↪conv4_block2_3_conv[0... (BatchNormalization)			└
↪			
conv4_block2_add (Add)	(None, 12, 12, 1024)	0	└
↪conv4_block1_out[0] [0... ↪conv4_block2_3_bn[0] [...			└
conv4_block2_out	(None, 12, 12, 1024)	0	└
↪conv4_block2_add[0] [0] (Activation)			└
↪			
conv4_block3_1_conv	(None, 12, 12, 256)	262,400	└
↪conv4_block2_out[0] [0] (Conv2D)			└
↪			
conv4_block3_1_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block3_1_conv[0... (BatchNormalization)			└
↪			
conv4_block3_1_relu	(None, 12, 12, 256)	0	└
↪conv4_block3_1_bn[0] [...			

(Activation)			└
↪			
conv4_block3_2_conv	(None, 12, 12, 256)	590,080	└
↪conv4_block3_1_relu[0...]			
(Conv2D)			└
↪			
conv4_block3_2_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block3_2_conv[0...]			
(BatchNormalization)			└
↪			
conv4_block3_2_relu	(None, 12, 12, 256)	0	└
↪conv4_block3_2_bn[0] [...]			
(Activation)			└
↪			
conv4_block3_3_conv	(None, 12, 12, 1024)	263,168	└
↪conv4_block3_2_relu[0...]			
(Conv2D)			└
↪			
conv4_block3_3_bn	(None, 12, 12, 1024)	4,096	└
↪conv4_block3_3_conv[0...]			
(BatchNormalization)			└
↪			
conv4_block3_add (Add)	(None, 12, 12, 1024)	0	└
↪conv4_block2_out[0] [0...]			
			└
↪conv4_block3_3_bn[0] [...]			
conv4_block3_out	(None, 12, 12, 1024)	0	└
↪conv4_block3_add[0] [0]			
(Activation)			└
↪			
conv4_block4_1_conv	(None, 12, 12, 256)	262,400	└
↪conv4_block3_out[0] [0]			
(Conv2D)			└
↪			
conv4_block4_1_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block4_1_conv[0...]			

(BatchNormalization)			└
↪			
conv4_block4_1_relu	(None, 12, 12, 256)	0	└
↪conv4_block4_1_bn[0][...]			
(Activation)			└
↪			
conv4_block4_2_conv	(None, 12, 12, 256)	590,080	└
↪conv4_block4_1_relu[0][...]			
(Conv2D)			└
↪			
conv4_block4_2_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block4_2_conv[0][...]			
(BatchNormalization)			└
↪			
conv4_block4_2_relu	(None, 12, 12, 256)	0	└
↪conv4_block4_2_bn[0][...]			
(Activation)			└
↪			
conv4_block4_3_conv	(None, 12, 12, 1024)	263,168	└
↪conv4_block4_2_relu[0][...]			
(Conv2D)			└
↪			
conv4_block4_3_bn	(None, 12, 12, 1024)	4,096	└
↪conv4_block4_3_conv[0][...]			
(BatchNormalization)			└
↪			
conv4_block4_add (Add)	(None, 12, 12, 1024)	0	└
↪conv4_block3_out[0][0][...]			
			└
↪conv4_block4_3_bn[0][...]			
conv4_block4_out	(None, 12, 12, 1024)	0	└
↪conv4_block4_add[0][0][...]			
(Activation)			└
↪			
conv4_block5_1_conv	(None, 12, 12, 256)	262,400	└
↪conv4_block4_out[0][0][...]			

(Conv2D)			└
↪			
conv4_block5_1_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block5_1_conv[0... (BatchNormalization)			└
↪			
conv4_block5_1_relu	(None, 12, 12, 256)	0	└
↪conv4_block5_1_bn[0] [... (Activation)			└
↪			
conv4_block5_2_conv	(None, 12, 12, 256)	590,080	└
↪conv4_block5_1_relu[0... (Conv2D)			└
↪			
conv4_block5_2_bn	(None, 12, 12, 256)	1,024	└
↪conv4_block5_2_conv[0... (BatchNormalization)			└
↪			
conv4_block5_2_relu	(None, 12, 12, 256)	0	└
↪conv4_block5_2_bn[0] [... (Activation)			└
↪			
conv4_block5_3_conv	(None, 12, 12, 1024)	263,168	└
↪conv4_block5_2_relu[0... (Conv2D)			└
↪			
conv4_block5_3_bn	(None, 12, 12, 1024)	4,096	└
↪conv4_block5_3_conv[0... (BatchNormalization)			└
↪			
conv4_block5_add (Add)	(None, 12, 12, 1024)	0	└
↪conv4_block4_out[0] [0...			└
↪conv4_block5_3_bn[0] [...			
conv4_block5_out	(None, 12, 12, 1024)	0	└
↪conv4_block5_add[0] [0]			

(Activation)			└
↳			
conv4_block6_1_conv	(None, 12, 12, 256)	262,400	└
↳conv4_block5_out[0][0]			
(Conv2D)			└
↳			
conv4_block6_1_bn	(None, 12, 12, 256)	1,024	└
↳conv4_block6_1_conv[0...]			
(BatchNormalization)			└
↳			
conv4_block6_1_relu	(None, 12, 12, 256)	0	└
↳conv4_block6_1_bn[0][...]			
(Activation)			└
↳			
conv4_block6_2_conv	(None, 12, 12, 256)	590,080	└
↳conv4_block6_1_relu[0...]			
(Conv2D)			└
↳			
conv4_block6_2_bn	(None, 12, 12, 256)	1,024	└
↳conv4_block6_2_conv[0...]			
(BatchNormalization)			└
↳			
conv4_block6_2_relu	(None, 12, 12, 256)	0	└
↳conv4_block6_2_bn[0][...]			
(Activation)			└
↳			
conv4_block6_3_conv	(None, 12, 12, 1024)	263,168	└
↳conv4_block6_2_relu[0...]			
(Conv2D)			└
↳			
conv4_block6_3_bn	(None, 12, 12, 1024)	4,096	└
↳conv4_block6_3_conv[0...]			
(BatchNormalization)			└
↳			
conv4_block6_add (Add)	(None, 12, 12, 1024)	0	└
↳conv4_block5_out[0][0...]			

↪conv4_block6_3_bn[0][...]			└
conv4_block6_out	(None, 12, 12, 1024)	0	└
↪conv4_block6_add[0][0]			
(Activation)			└
↪			
conv5_block1_1_conv	(None, 6, 6, 512)	524,800	└
↪conv4_block6_out[0][0]			
(Conv2D)			└
↪			
conv5_block1_1_bn	(None, 6, 6, 512)	2,048	└
↪conv5_block1_1_conv[0...			
(BatchNormalization)			└
↪			
conv5_block1_1_relu	(None, 6, 6, 512)	0	└
↪conv5_block1_1_bn[0][...]			
(Activation)			└
↪			
conv5_block1_2_conv	(None, 6, 6, 512)	2,359,808	└
↪conv5_block1_1_relu[0...			
(Conv2D)			└
↪			
conv5_block1_2_bn	(None, 6, 6, 512)	2,048	└
↪conv5_block1_2_conv[0...			
(BatchNormalization)			└
↪			
conv5_block1_2_relu	(None, 6, 6, 512)	0	└
↪conv5_block1_2_bn[0][...]			
(Activation)			└
↪			
conv5_block1_0_conv	(None, 6, 6, 2048)	2,099,200	└
↪conv4_block6_out[0][0]			
(Conv2D)			└
↪			
conv5_block1_3_conv	(None, 6, 6, 2048)	1,050,624	└
↪conv5_block1_2_relu[0...			

(Conv2D)			⌞
↪			
conv5_block1_0_bn	(None, 6, 6, 2048)	8,192	⌞
↪conv5_block1_0_conv[0... (BatchNormalization)			⌞
↪			
conv5_block1_3_bn	(None, 6, 6, 2048)	8,192	⌞
↪conv5_block1_3_conv[0... (BatchNormalization)			⌞
↪			
conv5_block1_add (Add)	(None, 6, 6, 2048)	0	⌞
↪conv5_block1_0_bn[0] [...]			⌞
↪conv5_block1_3_bn[0] [...]			⌞
conv5_block1_out	(None, 6, 6, 2048)	0	⌞
↪conv5_block1_add[0] [0] (Activation)			⌞
↪			
conv5_block2_1_conv	(None, 6, 6, 512)	1,049,088	⌞
↪conv5_block1_out[0] [0] (Conv2D)			⌞
↪			
conv5_block2_1_bn	(None, 6, 6, 512)	2,048	⌞
↪conv5_block2_1_conv[0... (BatchNormalization)			⌞
↪			
conv5_block2_1_relu	(None, 6, 6, 512)	0	⌞
↪conv5_block2_1_bn[0] [...] (Activation)			⌞
↪			
conv5_block2_2_conv	(None, 6, 6, 512)	2,359,808	⌞
↪conv5_block2_1_relu[0... (Conv2D)			⌞
↪			
conv5_block2_2_bn	(None, 6, 6, 512)	2,048	⌞
↪conv5_block2_2_conv[0...			

(BatchNormalization)			└
↪			
conv5_block2_2_relu	(None, 6, 6, 512)	0	└
↪conv5_block2_2_bn[0][...]			
(Activation)			└
↪			
conv5_block2_3_conv	(None, 6, 6, 2048)	1,050,624	└
↪conv5_block2_2_relu[0]...			
(Conv2D)			└
↪			
conv5_block2_3_bn	(None, 6, 6, 2048)	8,192	└
↪conv5_block2_3_conv[0]...			
(BatchNormalization)			└
↪			
conv5_block2_add (Add)	(None, 6, 6, 2048)	0	└
↪conv5_block1_out[0][0]...			
			└
↪conv5_block2_3_bn[0][...]			
conv5_block2_out	(None, 6, 6, 2048)	0	└
↪conv5_block2_add[0][0]			
(Activation)			└
↪			
conv5_block3_1_conv	(None, 6, 6, 512)	1,049,088	└
↪conv5_block2_out[0][0]			
(Conv2D)			└
↪			
conv5_block3_1_bn	(None, 6, 6, 512)	2,048	└
↪conv5_block3_1_conv[0]...			
(BatchNormalization)			└
↪			
conv5_block3_1_relu	(None, 6, 6, 512)	0	└
↪conv5_block3_1_bn[0][...]			
(Activation)			└
↪			
conv5_block3_2_conv	(None, 6, 6, 512)	2,359,808	└
↪conv5_block3_1_relu[0]...			

(Conv2D)			⌵
↪			
conv5_block3_2_bn	(None, 6, 6, 512)	2,048	⌵
↪conv5_block3_2_conv[0...			
(BatchNormalization)			⌵
↪			
conv5_block3_2_relu	(None, 6, 6, 512)	0	⌵
↪conv5_block3_2_bn[0] [...]			
(Activation)			⌵
↪			
conv5_block3_3_conv	(None, 6, 6, 2048)	1,050,624	⌵
↪conv5_block3_2_relu[0...			
(Conv2D)			⌵
↪			
conv5_block3_3_bn	(None, 6, 6, 2048)	8,192	⌵
↪conv5_block3_3_conv[0...			
(BatchNormalization)			⌵
↪			
conv5_block3_add (Add)	(None, 6, 6, 2048)	0	⌵
↪conv5_block2_out[0] [0...			
			⌵
↪conv5_block3_3_bn[0] [...]			
conv5_block3_out	(None, 6, 6, 2048)	0	⌵
↪conv5_block3_add[0] [0]			
(Activation)			⌵
↪			

Total params: 23,587,712 (89.98 MB)

Trainable params: 23,534,592 (89.78 MB)

Non-trainable params: 53,120 (207.50 KB)

Fast feature extraction without data augmentation **Extracting the VGG16 features and corresponding labels**

```
[11]: import numpy as np

def get_features_and_labels(dataset):
    all_features = []
    all_labels = []
    for images, labels in dataset:
        preprocessed_images = keras.applications.resnet.preprocess_input(images)
        features = conv_base.predict(preprocessed_images)
        all_features.append(features)
        all_labels.append(labels)
    return np.concatenate(all_features), np.concatenate(all_labels)
```

```
[12]: from keras.utils import image_dataset_from_directory

train_dataset = image_dataset_from_directory(
    new_base_dir / "train",
    image_size=(180, 180),
    batch_size=32)
validation_dataset = image_dataset_from_directory(
    new_base_dir / "validation",
    image_size=(180, 180),
    batch_size=32)
test_dataset = image_dataset_from_directory(
    new_base_dir / "test",
    image_size=(180, 180),
    batch_size=32)
```

Found 2000 files belonging to 2 classes.

Found 1000 files belonging to 2 classes.

Found 2000 files belonging to 2 classes.

```
[13]: train_features, train_labels = get_features_and_labels(train_dataset)
val_features, val_labels = get_features_and_labels(validation_dataset)
test_features, test_labels = get_features_and_labels(test_dataset)
```

```
1/1          5s 5s/step
1/1          3s 3s/step
1/1          4s 4s/step
1/1          3s 3s/step
1/1          3s 3s/step
1/1          5s 5s/step
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1/1          3s 3s/step
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```

1/1	3s 3s/step
1/1	5s 5s/step
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1/1	3s 3s/step
1/1	3s 3s/step
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1/1	5s 5s/step
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1/1	5s 5s/step
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1/1	4s 4s/step
1/1	3s 3s/step
1/1	3s 3s/step
1/1	5s 5s/step
1/1	3s 3s/step
1/1	3s 3s/step
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1/1	3s 3s/step
1/1	4s 4s/step
1/1	4s 4s/step
1/1	3s 3s/step
1/1	3s 3s/step
1/1	5s 5s/step
1/1	3s 3s/step
1/1	3s 3s/step

1/1 2s 2s/step

```
[14]: random_numbers = np.random.normal(size=(1000, 16))
dataset = tf.data.Dataset.from_tensor_slices(random_numbers)
```

```
[15]: for i, element in enumerate(dataset):
        print(element.shape)
        if i >= 2:
            break
```

(16,)

(16,)

(16,)

```
[16]: batched_dataset = dataset.batch(32)
for i, element in enumerate(batched_dataset):
    print(element.shape)
    if i >= 2:
        break
```

(32, 16)

(32, 16)

(32, 16)

```
[17]: reshaped_dataset = dataset.map(lambda x: tf.reshape(x, (4, 4)))
for i, element in enumerate(reshaped_dataset):
    print(element.shape)
    if i >= 2:
        break
```

(4, 4)

(4, 4)

(4, 4)

Displaying the shapes of the data and labels yielded by the Dataset

```
[18]: for data_batch, labels_batch in train_dataset:
        print("data batch shape:", data_batch.shape)
        print("labels batch shape:", labels_batch.shape)
        break
```

data batch shape: (32, 180, 180, 3)

labels batch shape: (32,)

Defining and training the densely connected classifier

```
[19]: data_augmentation = keras.Sequential(
    [
        layers.RandomFlip("horizontal"),
        layers.RandomRotation(0.1),
        layers.RandomZoom(0.2),
```

```

    ]
)

inputs = keras.Input(shape=(180, 180, 3))
x = data_augmentation(inputs)
x = keras.applications.resnet.preprocess_input(x)
x = conv_base(x)
x = layers.Flatten()(x)
x = layers.Dense(256)(x)
x = layers.Dropout(0.5)(x)
outputs = layers.Dense(1, activation="sigmoid")(x)
model = keras.Model(inputs, outputs)
conv_base.trainable = False

model.compile(loss="binary_crossentropy",
              optimizer="rmsprop",
              metrics=["accuracy"])

```

```

[20]: callbacks = [
        keras.callbacks.ModelCheckpoint(
            filepath="resnet_from_scratch.keras",
            save_best_only=True,
            monitor="val_loss")
    ]
history = model.fit(
    train_dataset,
    epochs=30,
    validation_data=validation_dataset,
    callbacks=callbacks)

```

```

Epoch 1/30
63/63          412s 6s/step -
accuracy: 0.8685 - loss: 21.1755 - val_accuracy: 0.9760 - val_loss: 3.1565
Epoch 2/30
63/63          438s 6s/step -
accuracy: 0.9316 - loss: 12.7385 - val_accuracy: 0.9830 - val_loss: 2.9085
Epoch 3/30
63/63          440s 6s/step -
accuracy: 0.9628 - loss: 6.3500 - val_accuracy: 0.9750 - val_loss: 4.7490
Epoch 4/30
63/63          400s 6s/step -
accuracy: 0.9423 - loss: 10.3162 - val_accuracy: 0.9680 - val_loss: 6.3492
Epoch 5/30
63/63          443s 6s/step -
accuracy: 0.9684 - loss: 5.0104 - val_accuracy: 0.9830 - val_loss: 3.2209
Epoch 6/30
63/63          440s 6s/step -
accuracy: 0.9727 - loss: 3.9360 - val_accuracy: 0.9630 - val_loss: 6.5376

```

Epoch 7/30
63/63 441s 6s/step -
accuracy: 0.9691 - loss: 3.8774 - val_accuracy: 0.9840 - val_loss: 3.0258
Epoch 8/30
63/63 445s 6s/step -
accuracy: 0.9792 - loss: 3.1603 - val_accuracy: 0.9770 - val_loss: 3.8296
Epoch 9/30
63/63 441s 6s/step -
accuracy: 0.9693 - loss: 2.7350 - val_accuracy: 0.9810 - val_loss: 3.8759
Epoch 10/30
63/63 441s 6s/step -
accuracy: 0.9855 - loss: 2.1458 - val_accuracy: 0.9570 - val_loss: 10.4142
Epoch 11/30
63/63 413s 6s/step -
accuracy: 0.9826 - loss: 2.0903 - val_accuracy: 0.9780 - val_loss: 3.7855
Epoch 12/30
63/63 371s 6s/step -
accuracy: 0.9832 - loss: 2.1665 - val_accuracy: 0.9760 - val_loss: 3.1822
Epoch 13/30
63/63 399s 6s/step -
accuracy: 0.9849 - loss: 1.3487 - val_accuracy: 0.9780 - val_loss: 3.3692
Epoch 14/30
63/63 401s 6s/step -
accuracy: 0.9876 - loss: 1.9007 - val_accuracy: 0.9770 - val_loss: 2.9153
Epoch 15/30
63/63 415s 6s/step -
accuracy: 0.9803 - loss: 2.3153 - val_accuracy: 0.9750 - val_loss: 3.9513
Epoch 16/30
63/63 384s 6s/step -
accuracy: 0.9770 - loss: 3.2521 - val_accuracy: 0.9780 - val_loss: 3.4706
Epoch 17/30
63/63 409s 6s/step -
accuracy: 0.9866 - loss: 1.4428 - val_accuracy: 0.9760 - val_loss: 3.8745
Epoch 18/30
63/63 439s 6s/step -
accuracy: 0.9915 - loss: 1.1439 - val_accuracy: 0.9780 - val_loss: 3.5105
Epoch 19/30
63/63 446s 6s/step -
accuracy: 0.9920 - loss: 1.1759 - val_accuracy: 0.9800 - val_loss: 2.3986
Epoch 20/30
63/63 438s 6s/step -
accuracy: 0.9788 - loss: 1.9644 - val_accuracy: 0.9820 - val_loss: 2.7674
Epoch 21/30
63/63 441s 6s/step -
accuracy: 0.9903 - loss: 1.0850 - val_accuracy: 0.9840 - val_loss: 2.9010
Epoch 22/30
63/63 413s 6s/step -
accuracy: 0.9851 - loss: 1.7150 - val_accuracy: 0.9820 - val_loss: 3.0053

```

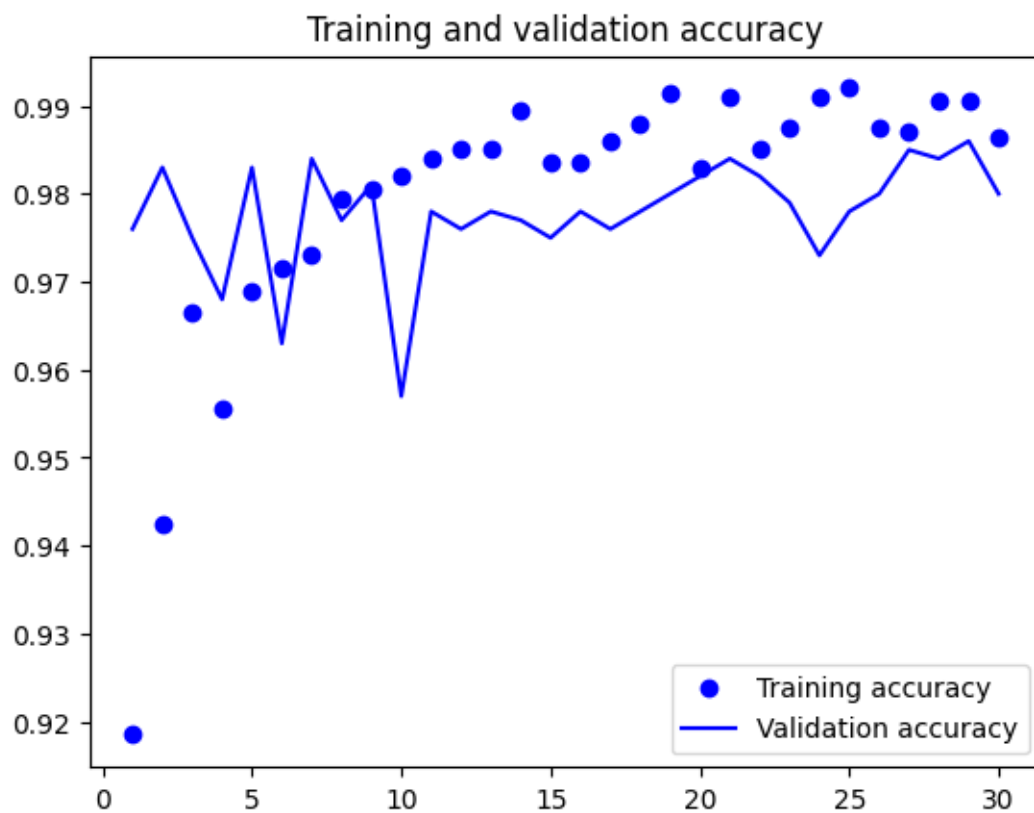
Epoch 23/30
63/63          398s 6s/step -
accuracy: 0.9860 - loss: 2.2158 - val_accuracy: 0.9790 - val_loss: 4.0467
Epoch 24/30
63/63          399s 6s/step -
accuracy: 0.9915 - loss: 0.8208 - val_accuracy: 0.9730 - val_loss: 5.1504
Epoch 25/30
63/63          399s 6s/step -
accuracy: 0.9884 - loss: 0.9231 - val_accuracy: 0.9780 - val_loss: 3.5578
Epoch 26/30
63/63          368s 6s/step -
accuracy: 0.9862 - loss: 1.5906 - val_accuracy: 0.9800 - val_loss: 3.0991
Epoch 27/30
63/63          413s 6s/step -
accuracy: 0.9869 - loss: 1.1333 - val_accuracy: 0.9850 - val_loss: 2.3382
Epoch 28/30
63/63          371s 6s/step -
accuracy: 0.9837 - loss: 1.2410 - val_accuracy: 0.9840 - val_loss: 2.9358
Epoch 29/30
63/63          409s 6s/step -
accuracy: 0.9882 - loss: 1.1326 - val_accuracy: 0.9860 - val_loss: 2.5033
Epoch 30/30
63/63          442s 6s/step -
accuracy: 0.9848 - loss: 2.0203 - val_accuracy: 0.9800 - val_loss: 3.1764

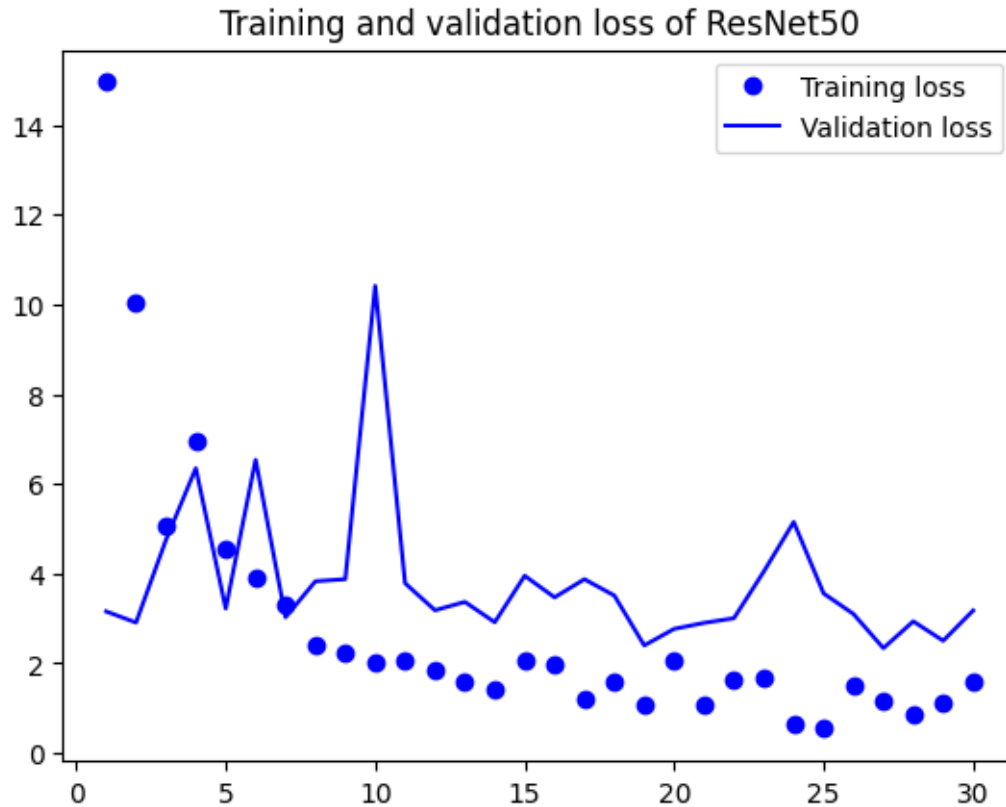
```

```

[21]: import matplotlib.pyplot as plt
acc = history.history["accuracy"]
val_acc = history.history["val_accuracy"]
loss = history.history["loss"]
val_loss = history.history["val_loss"]
epochs = range(1, len(acc) + 1)
plt.plot(epochs, acc, "bo", label="Training accuracy")
plt.plot(epochs, val_acc, "b", label="Validation accuracy")
plt.title("Training and validation accuracy")
plt.legend()
plt.figure()
plt.plot(epochs, loss, "bo", label="Training loss")
plt.plot(epochs, val_loss, "b", label="Validation loss")
plt.title("Training and validation loss of ResNet50")
plt.legend()
plt.show()

```





```
[22]: model = keras.models.load_model("resnet_from_scratch.keras")
      test_loss, test_acc = model.evaluate(test_dataset)
      print(f"Test accuracy ResNet50: {test_acc:.3f}")
```

```
63/63          231s 4s/step -
accuracy: 0.9749 - loss: 5.0084
Test accuracy ResNet50: 0.979
```

```
[23]: conv_base.trainable = True
      for layer in conv_base.layers[:-4]:
          layer.trainable = False
```

Fine-tuning the model

```
[24]: model.compile(loss="binary_crossentropy",
                    optimizer=keras.optimizers.RMSprop(learning_rate=1e-5),
                    metrics=["accuracy"])

callbacks = [
    keras.callbacks.ModelCheckpoint(
        filepath="fine_tuning.keras",
        save_best_only=True,
```

```

        monitor="val_loss")
]
history = model.fit(
    train_dataset,
    epochs=30,
    validation_data=validation_dataset,
    callbacks=callbacks)

```

Epoch 1/30

63/63 414s 6s/step -
accuracy: 0.9887 - loss: 0.8430 - val_accuracy: 0.9840 - val_loss: 2.3098

Epoch 2/30

63/63 405s 6s/step -
accuracy: 0.9915 - loss: 0.8798 - val_accuracy: 0.9840 - val_loss: 2.2884

Epoch 3/30

63/63 443s 6s/step -
accuracy: 0.9906 - loss: 1.1612 - val_accuracy: 0.9840 - val_loss: 2.2781

Epoch 4/30

63/63 408s 7s/step -
accuracy: 0.9899 - loss: 1.0235 - val_accuracy: 0.9830 - val_loss: 2.2534

Epoch 5/30

63/63 433s 6s/step -
accuracy: 0.9900 - loss: 0.5970 - val_accuracy: 0.9830 - val_loss: 2.2541

Epoch 6/30

63/63 450s 7s/step -
accuracy: 0.9875 - loss: 0.7755 - val_accuracy: 0.9830 - val_loss: 2.2514

Epoch 7/30

63/63 445s 7s/step -
accuracy: 0.9909 - loss: 0.5419 - val_accuracy: 0.9840 - val_loss: 2.2468

Epoch 8/30

63/63 432s 6s/step -
accuracy: 0.9916 - loss: 0.4181 - val_accuracy: 0.9840 - val_loss: 2.2509

Epoch 9/30

63/63 424s 6s/step -
accuracy: 0.9860 - loss: 1.6898 - val_accuracy: 0.9840 - val_loss: 2.2451

Epoch 10/30

63/63 439s 6s/step -
accuracy: 0.9928 - loss: 0.6897 - val_accuracy: 0.9840 - val_loss: 2.2422

Epoch 11/30

63/63 408s 6s/step -
accuracy: 0.9857 - loss: 1.0778 - val_accuracy: 0.9840 - val_loss: 2.2401

Epoch 12/30

63/63 443s 6s/step -
accuracy: 0.9915 - loss: 0.4185 - val_accuracy: 0.9840 - val_loss: 2.2364

Epoch 13/30

63/63 433s 6s/step -
accuracy: 0.9934 - loss: 0.5567 - val_accuracy: 0.9840 - val_loss: 2.2370

Epoch 14/30

63/63 443s 6s/step -
accuracy: 0.9955 - loss: 0.2808 - val_accuracy: 0.9830 - val_loss: 2.2429
Epoch 15/30

63/63 443s 6s/step -
accuracy: 0.9960 - loss: 0.3391 - val_accuracy: 0.9850 - val_loss: 2.2593
Epoch 16/30

63/63 444s 6s/step -
accuracy: 0.9844 - loss: 1.0722 - val_accuracy: 0.9850 - val_loss: 2.2743
Epoch 17/30

63/63 438s 6s/step -
accuracy: 0.9889 - loss: 0.8241 - val_accuracy: 0.9850 - val_loss: 2.2673
Epoch 18/30

63/63 442s 6s/step -
accuracy: 0.9941 - loss: 0.6597 - val_accuracy: 0.9850 - val_loss: 2.2784
Epoch 19/30

63/63 411s 6s/step -
accuracy: 0.9881 - loss: 1.6396 - val_accuracy: 0.9850 - val_loss: 2.2643
Epoch 20/30

63/63 417s 6s/step -
accuracy: 0.9910 - loss: 0.6492 - val_accuracy: 0.9840 - val_loss: 2.2318
Epoch 21/30

63/63 447s 7s/step -
accuracy: 0.9905 - loss: 0.9546 - val_accuracy: 0.9830 - val_loss: 2.2201
Epoch 22/30

63/63 413s 7s/step -
accuracy: 0.9877 - loss: 1.0318 - val_accuracy: 0.9830 - val_loss: 2.2146
Epoch 23/30

63/63 428s 6s/step -
accuracy: 0.9921 - loss: 0.7612 - val_accuracy: 0.9830 - val_loss: 2.2156
Epoch 24/30

63/63 442s 6s/step -
accuracy: 0.9923 - loss: 0.4031 - val_accuracy: 0.9850 - val_loss: 2.2416
Epoch 25/30

63/63 416s 6s/step -
accuracy: 0.9928 - loss: 1.0069 - val_accuracy: 0.9850 - val_loss: 2.2887
Epoch 26/30

63/63 409s 6s/step -
accuracy: 0.9936 - loss: 0.4542 - val_accuracy: 0.9860 - val_loss: 2.3078
Epoch 27/30

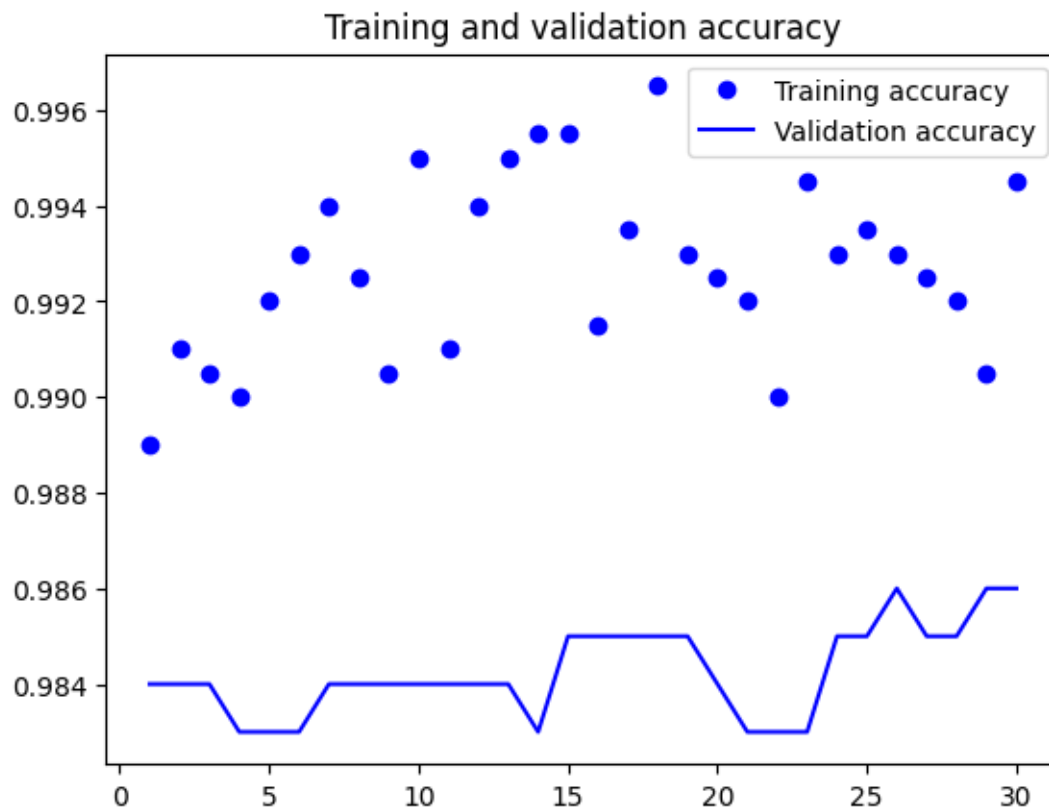
63/63 439s 6s/step -
accuracy: 0.9919 - loss: 0.3984 - val_accuracy: 0.9850 - val_loss: 2.3062
Epoch 28/30

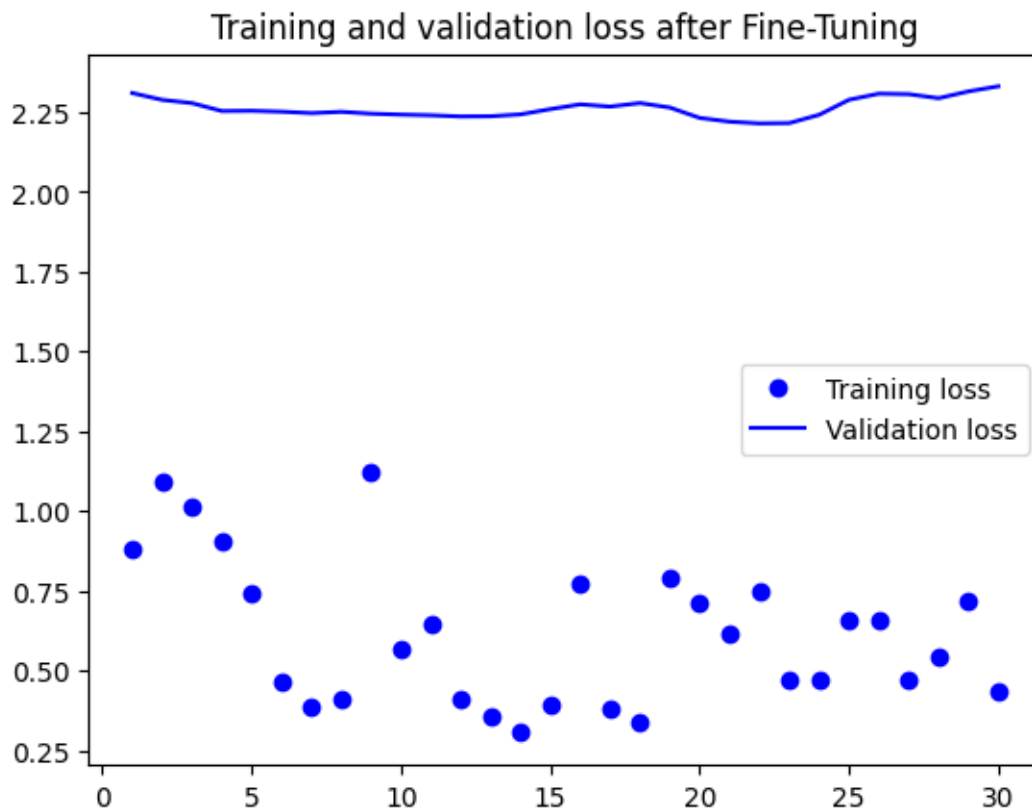
63/63 446s 6s/step -
accuracy: 0.9900 - loss: 0.7316 - val_accuracy: 0.9850 - val_loss: 2.2935
Epoch 29/30

63/63 438s 6s/step -
accuracy: 0.9898 - loss: 0.8779 - val_accuracy: 0.9860 - val_loss: 2.3150
Epoch 30/30

63/63 420s 6s/step -
accuracy: 0.9919 - loss: 0.7358 - val_accuracy: 0.9860 - val_loss: 2.3310

```
[25]: import matplotlib.pyplot as plt
acc = history.history["accuracy"]
val_acc = history.history["val_accuracy"]
loss = history.history["loss"]
val_loss = history.history["val_loss"]
epochs = range(1, len(acc) + 1)
plt.plot(epochs, acc, "bo", label="Training accuracy")
plt.plot(epochs, val_acc, "b", label="Validation accuracy")
plt.title("Training and validation accuracy")
plt.legend()
plt.figure()
plt.plot(epochs, loss, "bo", label="Training loss")
plt.plot(epochs, val_loss, "b", label="Validation loss")
plt.title("Training and validation loss after Fine-Tuning")
plt.legend()
plt.show()
```





```
[26]: model = keras.models.load_model("fine_tuning.keras")
      test_loss, test_acc = model.evaluate(test_dataset)
      print(f"Test accuracy after fine-tuning: {test_acc:.3f}")
```

```
63/63          235s 4s/step -
accuracy: 0.9830 - loss: 3.1652
Test accuracy after fine-tuning: 0.981
```